COURSE SYLLABUS

Web Development

23 modules | 92 sessions | live classes, recordings, assignments and placement support

01. Orientation and setup (Week 1)

- How the course works: classes, recordings and assignments
- Installing VS Code, Chrome and Git
- Folder structure and terminal basics
- Publishing your first page on the internet

02. How the web works (Week 1)

- Browsers, servers, HTTP and HTTPS
- Domains, hosting and DNS records
- Inspecting network requests in DevTools
- Static sites vs dynamic sites

03. HTML structure and semantics (Week 2)

- Document skeleton and the head section
- Headings, lists, links and images
- Semantic layout tags and when to use them
- Comments, indentation and readable markup

04. HTML forms and media (Week 2)

- Input types, labels, selects and textareas
- Validation attributes and error messages
- Embedding video, audio and maps
- Lab: contact and enquiry forms

05. CSS fundamentals (Week 3)

- Selectors, cascade and specificity
- The box model and spacing systems
- Typography, colour and custom properties
- Pseudo classes and hover states

06. Flexbox and grid (Week 3)

- Flex direction, wrapping and alignment
- Grid templates and gaps
- Sticky headers and sidebars
- Lab: card grid and pricing layout

07. Responsive and mobile first design (Week 4)

- Breakpoints and fluid type
- Responsive images and media queries
- Touch targets and mobile navigation
- Testing on real screen sizes

08. Modern CSS features (Week 4)

- Transitions, transforms and keyframes
- Dark mode and prefers-color-scheme
- Container queries and clamp()
- Reduced motion and accessibility

09. Accessibility essentials (Week 5)

- Semantic HTML as the first accessibility tool
- Keyboard navigation and focus order

- Contrast, alt text and form labels
- Screen reader test walkthrough

10. JavaScript basics (Week 5)

- Variables, types and operators
- Conditionals, loops and functions
- Arrays and objects
- Debugging in the console

11. DOM and events (Week 6)

- Selecting and updating elements
- Event listeners, bubbling and delegation
- Form handling and validation in JS
- Building a tabs and accordion component

12. Async JavaScript and APIs (Week 6)

- Promises and async await
- fetch, JSON and error handling
- Loading and empty states
- Lab: live weather or news widget

13. Git and GitHub workflow (Week 7)

- Commits, branches and merges
- Pull requests and code review
- Hosting with GitHub Pages
- Writing a project README

14. Tailwind CSS (Week 7)

- Utility first styling and configuration
- Responsive and state variants
- Reusable components with classes
- Rebuilding a landing page in Tailwind

15. React foundations (Week 8)

- Components, JSX and props
- State with useState
- Lists, keys and conditional rendering
- Splitting a design into components

16. React forms and data (Week 9)

- Controlled inputs and validation
- Effects and cleanup
- Fetching data and showing states
- Routing between pages

17. Next.js and SEO (Week 10)

- Pages, layouts and routing
- Metadata, Open Graph and sitemaps
- Image and font optimisation
- Deploying to Vercel

18. Performance and Core Web Vitals (Week 10)

- Measuring LCP, CLS and INP
- Image, font and script budgets
- Lazy loading and code splitting
- Lighthouse audits and fixes

19. Testing and quality (Week 11)

- Accessibility and HTML validation checks
- Unit testing small functions
- Cross browser testing checklist
- Bug triage and regression notes

20. Portfolio project 1: business website (Week 11)

- Brief, wireframe and content plan
- Build responsive pages with a design system
- Contact form and WhatsApp integration
- Launch, analytics and handover

21. Portfolio project 2: web app (Week 12)

- Feature planning and data shape
- CRUD with a hosted backend
- Auth and protected screens
- Deploy, test and demo

22. Freelance and client workflow (Week 12)

- Scoping work and writing proposals
- Pricing, milestones and invoices
- Communication and revision limits
- Handover docs and maintenance plans

23. Interview preparation (Week 13)

- HTML, CSS and JS interview questions
- Live coding on common UI tasks
- Explaining your projects clearly
- Mock interview and feedback

Fees, batch dates and mode (online / classroom): confirm the current numbers on the website or with the admissions team before enrolling. Yantrix Technology, Pune.